

MEDIATEK

MT8676 Yocto SDCard User Manual

Version: 1.0
Release date: 2024-08-12

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Version History

Version	Date	Author	Description
1.0	2024-08-12	Andy-Id Lu	Official release

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1 SDCard

1.1 Overview

1.1.1 Brief Introduction

This chapter introduces the hardware features, software configuration and function of SDCard controller, also contains the debugging methods for common problems.

1.1.2 Abbreviations

Table 1-1. Abbreviations

Abbreviation	Explanation
DDR50	Double Data Rate up to 50MB/s@50MHz
SDR104	Signal Data Rate up to 104MB/s@208MHz
SDR12	Signal Data Rate up to 12.5MB/s@25MHz
SDR25	Signal Data Rate up to 25MB/s@50MHz
SDR50	Signal Data Rate up to 50MB/s@100MHz
UHS-I	Ultra High Speed Phase I card

1.2 Architecture/Process Overview

1.2.1 SDCard Introduction

SDCard is a new generation of high-speed storage device based on semiconductor flash memory and developed from MultiMedia Card (MMC) format. It has high memory capacity, fast data transfer rate, great mobility flexibility, and good security, which is widely used on portable devices. In the SD3.0 protocol, theoretical maximum capacity of SDCard could be up to 2TB, the theoretical maximum speed of read/write could be up to 104MB/s.

The major pins and functions are described as below:

- (1) CLK: Clock signal, host controller or SDCard transmit one cmd/data bit at each clock cycle, up to 208MHz in UHS-I speed mode;
- (2) CMD: Command and response shared pin, command is transmitted from host controller to SDCard, and response is from SDCard to host controller;
- (3) DAT0~3: Data lines, data could be transmitted from SDCard to host controller(read), and could also be transmitted from host controller to SDCard (write);
- (4) VDD: Power supply pin of SDCard, which is normally configured to 3.3V voltage, the scope specified in the SD protocol is 2.7V~3.6V;
- (5) CD: Detect pin of SDCard, normally achieve GPIO level variation pin in the state of card inserting and removing through the mechanical structure of SDCard holder.

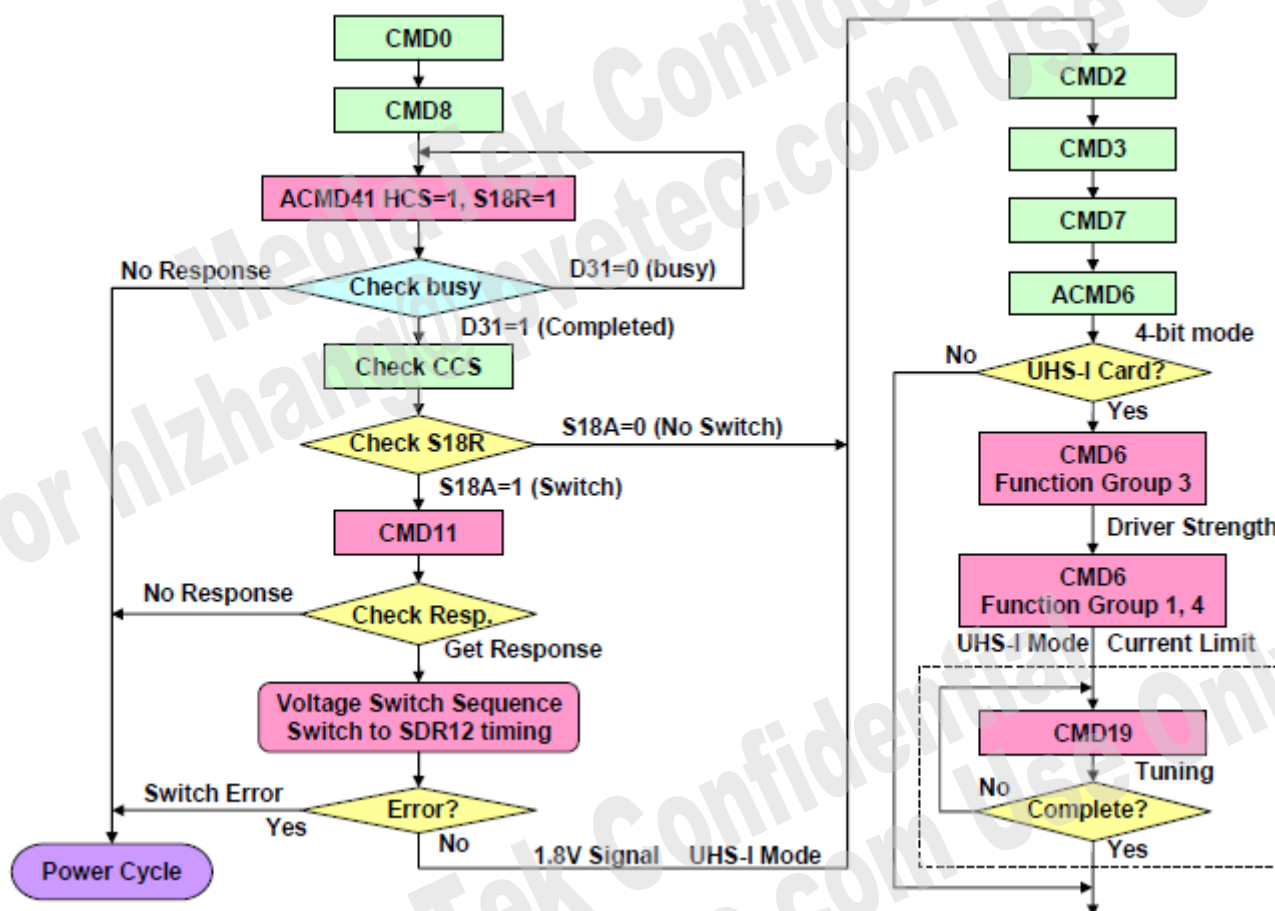


Figure 1-1. Initialization flow of UHS-I card

1.2.2 MT8676 SDCard Feature

- (1) Compatible with SD3.0 protocol standards
- (2) Supports Basic DMA and Descriptor DMA mode
- (3) Supports Bus speed mode: Default Speed/High Speed/SDR12/SDR25/SDR50/SDR104/DDR50
- (4) Supports 1/4bits bus width

1.3 Configuration/Customization Guideline

1.3.1 Kernel Config

- (1) Enables SDCard support
CONFIG_MMC = y
- (2) Enables Mediatek host driver support
CONFIG_MMC_MTK_PRO = m

1.3.2 DTS Node

```

10 &mmc1 {
11     host-index = <1>;
12     status = "okay";
13     pinctrl-names = "default", "state_uhs";
14     pinctrl-0 = <&mmc1_pins_default>;
15     pinctrl-1 = <&mmc1_pins_uhs>;
16     bus-width = <4>;
17     max-frequency = <200000000>;
18     ocr-voltage = <0x00070000>;
19     cap-sd-highspeed;
20     sd-uhs-sdr12;
21     sd-uhs-sdr25;
22     sd-uhs-sdr50;
23     sd-uhs-sdr104;
24     sd-uhs-ddr50;
25     cd-debounce-delay-ms = <0>;
26     cd-gpios = <&pio 33 GPIO_ACTIVE_LOW>;
27     vmmc-supply = <&mt6373_vmch_eint_high>;
28     vqmmc-supply = <&mt6373_vmc>;
29     no-mmc;
30     no-sdio;
31 };

```

Figure 1-2. DTS node for SDCard

- (1) SD2.0 card needs to configure "cap-sd-highspeed", SD3.0 ultra-high speed card needs to configure "sd-uhs-xxx";
- (2) SD driving strength could be configured in the pinctrl node of the corresponding mode, such as below SDR104 mode, such as the SDR104 mode shown as Figure 1-3;

```

74 mmc1_pins_uhs: mmc1@0{
75     pins-cmd-dat {
76         pinmux = <PINMUX_GPIO74_FUNC_MSDC1_DAT0>,
77                 <PINMUX_GPIO75_FUNC_MSDC1_DAT1>,
78                 <PINMUX_GPIO76_FUNC_MSDC1_DAT2>,
79                 <PINMUX_GPIO77_FUNC_MSDC1_DAT3>,
80                 <PINMUX_GPIO73_FUNC_MSDC1_CMD>;
81         input-enable;
82         drive-strength = <3>;
83         bias-pull-up = <MTK_PUPD_SET_R1R0_01>;
84     };
85
86     pins-clk {
87         pinmux = <PINMUX_GPIO72_FUNC_MSDC1_CLK>;
88         drive-strength = <3>;
89         bias-pull-down = <MTK_PUPD_SET_R1R0_10>;
90     };
91 };

```

Figure 1-3. Pinctrl node of SDR104 mode

- (3) SDCard detect pin could be configured by "cd-gpios", GPIO_ACTIVE_LOW represent low card insertion level, and GPIO_ACTIVE_HIGH represent high card insertion level;
- (4) The "vmmc-supply" and "vqmmc-supply" are configured according to actual used SDCard VDD power and Host I/O power. If it is necessary to use fast-power-off (VMCH hardware power off when SDCard is pulled out) function, "vmmc-supply" should be configured as &mt6373_vmch_eint_high (match with GPIO_ACTIVE_LOW of "cd-gpios") or &mt6373_vmch_low (match with GPIO_ACTIVE_HIGH of "cd-gpios"); and if it is unnecessary to use fast-power-off function, "vmmc-supply" should be configured as &mt6373_vmch.

1.3.3 KO Table

Adds host driver ko to the following ko table path, "ramdisk" is configured in the column three, which means that ko would be installed into initramfs.

meta/meta-mediatek-mt8676/recipes-kernel/linux/ko_order_table/\${PROJECT}/ko_order_table.csv:

```
147 mtk-mmc.ko,../../kernel_device_modules-6.1/drivers/mmc/host/mtk-mmc.ko,ramdisk,Y,Y,user/userdebug/eng
```

1.4 Frequently Asked Questions/Troubleshooting

1.4.1 SDCard is not Recognized, VDD Voltage could not be Measured

- (1) Checks the correction of Kernel config and DTS according to the previous section;
- (2) If VDD supplied power is MT6373, and detect pin is connected to SD_DET pin of MT6373, check if the configured power node of "vmmc-supply" is matched with the polarity of detect pin;
- (3) If the result of step (2) is okay, configure "vmmc-supply" as &mt6373_vmch and check VDD voltage, correct voltage represents there is problem in fast-power-off function, submit PMIC issue to MTK;
- (4) If VDD could not power up yet in step (3), catch kernel log and submit SDCard issue to MTK.

1.4.2 SDCard is Recognized while Power On with Card but Failed while Hot-plug

- (1) Checks the correction of "cd-gpios" configuration in DTS according to the previous section;
- (2) If DTS configuration is correct, check the correction of the GPIO configuration for detect pin in the file whose path is src/devtools/dct/dws/mt6897/\${PROJECT}.dws, refer to the configuration shown in [Figure 1-4](#);

```
430 <gpio33>
431   <eint_mode>true</eint_mode>
432   <def_mode>0</def_mode>
433   <inpull_en>true</inpull_en>
434   <inpull_selhigh>true</inpull_selhigh>
435   <def_dir>IN</def_dir>
436   <out_high>>false</out_high>
437   <varName0>GPIO_SDHC_EINT_PIN</varName0>
438   <smt>true</smt>
439   <ies>true</ies>
440 </gpio33>
```

Figure 1-4. dws setting for detect pin

- (3) If the DTS and dws configuration are all correct but SDCard could not be recognized yet while hot-plug, measure the level of detect pin in the state of card inserting and removing, submit SDCard issue to MTK if the result of measurement meets expectations.

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